

OMAR TAWFIK

Data Scientist, Causal Inference, Forecasting & Applied ML
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PROFESSIONAL SUMMARY

Data Scientist with 4+ years building reproducible forecasting, predictive modeling, and analytics solutions for high-stakes decisions. Experienced in Python, SQL, feature engineering, time-series analysis, regression, ETL pipelines, and executive-facing visualization, including probabilistic forecasting models that quantified \$31M in settlement exposure. Also develops applied AI systems and cloud-based data workflows that improve reporting efficiency and support scalable decision-making.

EXPERIENCE

Office of Labor Relations & Collective Bargaining (OLRCB)

2026 - Present

Data Scientist

- Quantified \$31M in potential settlement liability for a 26,000+ person workforce by building a compensation and back-pay prediction pipeline on structured payroll and demographic data, applying feature engineering, regression, and cross-validated statistical modeling to produce auditable, reproducible outputs.
- Replaced single-point fiscal estimates with probabilistic exposure ranges (P5-P95) by engineering multi-year forecasting models across 5 negotiation strategies, combining time-series forecasting, regression, and stochastic sensitivity analysis to support executive decision-making.
- Directly informed negotiated outcomes for a 26,000+ employee workforce through labor economics and compensation analyses across 10+ union matters, using Python and SQL.
- Eliminated manual tracking gaps across 20+ compliance obligations (CBAs, arbitration awards, PERB decisions) by architecting a centralized analytics and compliance framework with SQL-based data models and ML Ops practices for automated, audit-ready workflows.
- Built an end-to-end causal inference platform (Python, Flask, SQL) combining three causal models (DiD, Synthetic Control, PSM) via inverse-variance weighting, validated with Monte Carlo simulations and Chart.js dashboards, to quantify municipal wage policy fiscal outcomes across P5-P95 confidence intervals.

Booz Allen Hamilton Inc.

Jul 2023 - Nov 2024

Data Scientist

- Architected an end-to-end ETL pipeline using SQL and Azure that replaced manual reporting workflows, lifting team productivity 10x and improving data accuracy 50% across 10+ biweekly executive reports over a 5-month cycle.
- Led a 6-person team building Tableau dashboards and KPIs with Python and Apache Spark, standardizing reporting definitions across 4 business units and translating large-scale data analyses into executive-facing insights.
- Redesigned Salesforce LTV tracking infrastructure, improving response time by 35% and scaling platform capacity to 5,000+ monthly active users.

PROJECTS

DocMind: RAG Document Intelligence Pipeline [Live Deployment](#)

Jan 2026 - Mar 2026

Independent

- Built and deployed a full retrieval-augmented generation (RAG) pipeline (Python, Flask, Render) that ingests PDF, TXT, and Markdown documents and answers user questions grounded in source material, with citations and relevance scores on every response.
- Engineered the retrieval layer using fastembed (ONNX) embeddings and a persistent, cloud-hosted Qdrant vector store, chunking documents with overlap and serving cosine-similarity search across a multi-document, multi-turn chat interface with a REST API (upload, chat, document management).
- Integrated an LLM (Groq-hosted Llama 3.3 70B) for grounded answer generation, with document-level filtering and duplicate-ingestion detection; validated retrieval quality with a 10-question evaluation harness, reaching 100% hit-rate@6 and 0.95 MRR.

ReviewGuard: Fraudulent Review Detection Engine

May 2024 - Aug 2024

Independent

- Engineered 18 behavioral and linguistic features, including VADER sentiment, lexical diversity, rating-sentiment mismatch, and reviewer behavior signals, then combined them with TF-IDF vectorization using 10,000 unigram and bigram features to detect fraudulent reviews across 42,000+ records.
- Benchmarked classification models including Logistic Regression, Random Forest, Linear SVM, and Gradient Boosting using stratified 5-fold cross-validation, achieving 90.68% accuracy and a 90.62 F1-score; presented findings to a panel of technical and executive leaders at a Booz Allen internal innovation conference.

SKILLS

AI & LLM Engineering: RAG, LLM APIs (Groq/Llama), Vector Databases (Qdrant), fastembed (ONNX) Embeddings, Prompt Engineering, Retrieval Evaluation (hit-rate, MRR), GenAI

Machine Learning & Statistics: Predictive Modeling, Feature Engineering, Statistical Modeling, Causal Inference (DiD, Synthetic Control, PSM), Time Series Forecasting, NLP & Text Classification, Experimental Design, Regression, Classification

Programming & Tools: Python, SQL, scikit-learn, Flask, Git/GitHub, ML Ops

Data Engineering & ETL: ETL Pipeline Development, Apache Spark, Data Cleansing, Data Flow Analysis, Databricks, Snowflake

Cloud & Visualization: Azure, Tableau, Power BI, Qlik, Chart.js, Supply Chain Planning

EDUCATION

Master of Science, Data Science

May 2026

University of Maryland, Smith School of Business

- GPA: 4.0

Bachelor of Science, Operations Mgmt. & Business Analytics

May 2023

University of Maryland

- GPA: 3.9